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United States Patent	12409903
Kind Code	B2
Date of Patent	September 09, 2025
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Part unit and structure for assembling vehicle body

Abstract

A part unit and a vehicle body assembly structure are disclosed. In particular, a part unit and a vehicle body assembly structure with improved workability of assembling parts into a vehicle body of a curved surface are disclosed.

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Appl. No.:	18/009950
Filed (or PCT Filed):	June 15, 2022
PCT No.:	PCT/KR2022/008483
PCT Pub. No.:	WO2023/158025
PCT Pub. Date:	August 24, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240199154 A1	Jun. 20, 2024

Foreign Application Priority Data

KR	10-2022-0019424	Feb. 15, 2022
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Publication Classification

Int. Cl.: B62D65/02 (20060101); F16B5/02 (20060101)

U.S. Cl.:

CPC B62D65/02 (20130101); F16B5/0233 (20130101);

Field of Classification Search

CPC: B62D (65/02); B62D (65/024); B62D (65/16); B62D (27/023); B62D (27/065); F16B (19/1072); F16B (21/02); F16B (5/0208); F16B (5/0233)

USPC: 296/146.1; 296/152; 296/187.01; 296/29

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Background/Summary

(1) This application is the national phase entry of international patent application no. PCT/KR2022/008483 filed Jun. 15, 2022 and claims the benefit of Korean patent application No.

10-2022-0019424, filed Feb. 15, 2022, the disclosures of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(2) The present disclosure relates to a part unit and a vehicle body assembly structure. In particular, the present disclosure relates to a part unit and a vehicle body assembly structure with improved workability of assembling parts into a vehicle body of a curved surface.

BACKGROUND ART

(3) A related art vehicle body assembly structure manages an appropriate dimensional tolerance of a vehicle body and vehicle parts when the vehicle body and the vehicle parts are assembled with each other.

(4) However, after fastening holes of the parts are inserted into a large number of bolts welded to a bracket serving as an inner plate of the vehicle body, a large number of nuts are fastened to the bolts with a tool.

(5) Therefore, since after a worker holds the part with one hand and temporarily fastens the large number of nuts to the bolts one by one with the other hand, the worker holds the tool with the other hand and performs final tightening, the worker is inconvenient due to a large number of works.

(6) Further, a process of welding the large number of bolts to the bracket has to be added separately.

(7) In addition, it is difficult to manufacture the bracket, and it is necessary to fix a large number of brackets to an outer plate of the vehicle body one by one.

DISCLOSURE OF INVENTION

Technical Problem

(8) An object of the present disclosure is to address the above-described and other problems.

(9) Another object of the present disclosure is to provide a part unit in which a fastening module with an improved tolerance including a bolt part and a nut part is assembled on a mounting part of a part.

(10) Another object of the present disclosure is to provide a simple vehicle body assembly structure in which a worker holds a part unit, in which a fastening module with an improved tolerance including a bolt part and a nut part is assembled on a mounting part of a part, with one hand and with the other hand and fastens and couples the part unit to a vehicle body by turning the bolt part with a tool with the other hand.

Solution to Problem

(11) In order to achieve the above-described and other objects, in one aspect of the present disclosure, there is provided a part unit assembled to a vehicle body part comprising a part on which a plurality of mounting parts is formed; and a fastening module assembled to each of the plurality of mounting parts, wherein the fastening module includes a screw bushing part installed on the mounting part to be linearly movable back and forth; a bolt part fitted toward a front from a rear of the screw bushing part; and a nut part fastened to a male screw rod of the bolt part protruding forward, wherein the nut part passes from an inside to an outside of the vehicle body part, and then the nut part and the screw bushing part closely couple inner and outer surfaces of the vehicle body part by a pressurization rotation of the bolt part.

(12) In order to achieve the above-described and other objects, in another aspect of the present disclosure, there is provided a vehicle body assembly structure in which a part unit is assembled to a vehicle body part, wherein the vehicle body part includes a vehicle body outer plate and a vehicle body inner plate, wherein the vehicle body inner plate includes an insertion part penetrated back and forth; and an anti-rotation piece formed on an outer surface of the vehicle body inner plate, wherein the part unit includes a part on which a plurality of mounting parts is formed; and a fastening module configured to fasten and couple the plurality of mounting parts to the vehicle body inner plate, wherein the fastening module includes a screw bushing part installed on the mounting part to be linearly movable back and forth; a bolt part fitted toward a front from a rear of

the screw bushing part; and a nut part fastened to a male screw rod of the bolt part protruding forward, wherein the nut part is inserted into the insertion part and is disposed between the vehicle body outer plate and the vehicle body inner plate, and wherein the nut part and the screw bushing part closely couple inner and outer surfaces of the vehicle body inner plate by a rotation of the bolt part.

Advantageous Effects of Invention

(13) Effects of a part unit and a vehicle body assembly structure according to the present disclosure are described as follows.

(14) According to at least one embodiment of the present disclosure, the present disclosure can provide a part unit implementing a fastening module in which a bolt and a nut are installed.

(15) According to at least one embodiment of the present disclosure, the present disclosure can provide a part unit with improved assemblability in which a fastening module is assembled to a part once.

(16) According to at least one embodiment of the present disclosure, the present disclosure can provide a vehicle body assembly structure that can be expected to improve quality and reduce the cost by improving a process and the assembly workability of a part unit to a vehicle body.

(17) Additional scope of applicability of the present disclosure will become apparent from the detailed description given below. However, it should be understood that the detailed description and specific examples such as embodiments of the present disclosure are given merely by way of example, since various changes and modifications within the spirit and scope of the present disclosure will become apparent to those skilled in the art from the detailed description.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIGS. 1 and 2 are coupling and separation perspective views of a vehicle body assembly structure according to an embodiment of the present disclosure when viewed from the front.

(2) FIGS. 3 and 4 are rear perspective views of FIGS. 1 and 2.

(3) FIGS. 5 and 6 are front and rear perspective views dividedly illustrating a fastening module.

(4) FIG. 7 is a perspective view before a fastening module is assembled to a mounting part of a part.

(5) FIGS. 8 and 9 are perspective views illustrating that a fastening module is inserted into and rotation-coupled to a mounting part of a part.

(6) FIG. 10 is a cross-sectional view taken along line A-A of FIG. 8.

(7) FIGS. 11 to 16 illustrate order of assembling parts of FIG. 1 to a vehicle body.

(8) FIG. 17 is a separation perspective view of a vehicle body assembly structure according to another embodiment of the present disclosure when viewed from the rear.

(9) FIG. 18 is a perspective view illustrating a fastening module of FIG. 17.

(10) FIGS. 19 to 22 illustrate order of assembling parts of FIG. 17 to a vehicle body.

(11) FIG. 23 is a separation perspective view of a vehicle body assembly structure according to yet another embodiment of the present disclosure when viewed from the rear.

(12) FIG. 24 is a perspective view illustrating a fastening module of FIG. 23.

(13) FIGS. 25 to 28 illustrate order of assembling parts of FIG. 23 to a vehicle body.

(14) FIGS. 29 and 30 are perspective views illustrating before and after a guide cover is assembled to a part.

(15) FIG. 31 is a perspective view illustrating coupling parts of FIG. 30 to a vehicle body.

(16) FIG. 32 is a cross-sectional view taken along line A-A of FIG. 31.

MODE FOR THE INVENTION

(17) Reference will now be made in detail to embodiments of the disclosure, examples of which are

illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts. In general, a suffix such as “module” and “unit” may be used to refer to elements or components. Use of such a suffix herein is merely intended to facilitate description of the present disclosure, and the suffix itself is not intended to give any special meaning or function. It will be noted that a detailed description of known arts will be omitted if it is determined that the detailed description of the known arts can obscure the embodiments of the disclosure. The accompanying drawings are used to help easily understand various technical features and it should be understood that embodiments presented herein are not limited by the accompanying drawings. As such, the present disclosure should be construed to extend to any alterations, equivalents and substitutes in addition to those which are particularly set out in the accompanying drawings.

(18) The terms including an ordinal number such as first, second, etc. may be used to describe various components, but the components are not limited by such terms. The terms are used only for the purpose of distinguishing one component from other components.

(19) When any component is described as “being connected” or “being coupled” to other component, this should be understood to mean that another component may exist between them, although any component may be directly connected or coupled to the other component. In contrast, when any component is described as “being directly connected” or “being directly coupled” to other component, this should be understood to mean that no component exists between them.

(20) A singular expression can include a plural expression as long as it does not have an apparently different meaning in context.

(21) In the present disclosure, terms “include” and “have” should be understood to be intended to designate that illustrated features, numbers, steps, operations, components, parts or combinations thereof are present and not to preclude the existence of one or more different features, numbers, steps, operations, components, parts or combinations thereof, or the possibility of the addition thereof.

(22) In the drawings, sizes of the components may be exaggerated or reduced for convenience of explanation. For example, the size and the thickness of each component illustrated in the drawings are arbitrarily illustrated for convenience of explanation, and thus the present disclosure is not limited thereto unless specified as such.

(23) If any embodiment is implementable differently, a specific order of processes may be performed differently from the order described. For example, two consecutively described processes may be performed substantially at the same time, or performed in the order opposite to the described order.

(24) In the following embodiments, when layers, areas, components, etc. are connected, the following embodiments include both the case where layers, areas, and components are directly connected, and the case where layers, areas, and components are indirectly connected with other layers, areas, and components intervening between them. For example, when layers, areas, components, etc. are electrically connected, the present disclosure includes both the case where layers, areas, and components are directly electrically connected, and the case where layers, areas, and components are indirectly electrically connected with other layers, areas, and components intervening between them.

(25) Hereinafter, an x-axis is defined as a left-right direction, y-axis is defined as an inside-outside direction or a front-rear direction, and a z-axis is defined as an up-down direction.

(26) FIGS. 1 and 2 are coupling and separation perspective views of a vehicle body assembly structure according to an embodiment of the present disclosure when viewed from the front. FIGS. 3 and 4 are rear perspective views of FIGS. 1 and 2.

(27) Referring to FIGS. 1 to 4, a vehicle body assembly structure according to an embodiment of the present disclosure may include a part unit 1.

(28) The part unit 1 may be assembled to a vehicle body part 30.

- (29) The part unit **1** may include a part **3**. The part **3** may be a flush door from which a grip is withdraw.
- (30) The part **3** may include a housing **5**. The housing **5** may be formed by assembling front and rear housings. A grip **4** may be disposed on a front surface of the housing **5**. An operating part (not shown) withdrawing the grip **4** may be mounted inside the housing **5**.
- (31) The part **3** may include a mounting part **7**. The mounting part **7** may be installed on upper and lower sides of the housing **5** or formed integrally. That is, the mounting part **7** may include two mounting parts at upper left and right sides and one mounting part at a lower side. Referring to FIG. 7, the mounting part **7** may have a ring shape having a through hole.
- (32) The part unit **1** may include a fastening module **10**. The fastening module **10** may be assembled to the mounting part **7**. The fastening module **10** may be coupled to a vehicle body.
- (33) FIGS. 5 and 6 are front and rear perspective views dividedly illustrating a fastening module.
- (34) Referring to FIGS. 5 and 6, the fastening module **10** may include a screw bushing part **100**. The screw bushing part **100** may be installed on the mounting part **7** to be linearly movable back and forth.
- (35) The screw bushing part **100** may include a screw cover part **110**. The screw cover part **110** may be installed on the mounting part **7**. The screw cover part **110** can improve the assemblability of the fastening module **10** to the mounting part **7**.
- (36) The screw cover part **110** may include a hollow cover **120**. An inner screw **121** may be formed on an inner circumferential surface of the hollow cover **120**.
- (37) The screw cover part **110** may include a detachable part **130**. The detachable part **130** may detach the hollow cover **120** from the mounting part **7**.
- (38) Referring to FIG. 7, the detachable part **130** may include a first detachable part **131**. The first detachable part **131** may be formed on the mounting part **7**.
- (39) The first detachable part **131** may include a first front catching inner jaw **132**. The first front catching inner jaw **132** may be formed to protrude toward the center at the front left and right sides of the mounting part **7**.
- (40) The first detachable part **131** may include a first rear catching inner jaw **133**. The first rear catching inner jaw **133** may be formed on the rear side of the mounting part **7** in a diagonal direction. The first rear catching inner jaw **133** may be disposed on the upper and lower sides at the left and right sides when viewed from the front or rear.
- (41) Referring to FIGS. 5 and 6, the detachable part **130** may include a second detachable part **134**. The second detachable part **134** may be formed in the hollow cover **120**.
- (42) The second detachable part **134** may include a second front catching outer jaw **135**. The second front catching outer jaw **135** may be formed to protrude toward the center at the front left and right sides of the hollow cover **120**. The second front catching outer jaw **135** may pass between the first front catching inner jaw **132** and the first rear catching inner jaw **133** in the up-down direction. The second front catching outer jaw **135** may be caught on the first front catching inner jaw **132**.
- (43) The second detachable part **134** may include a second rear catching outer jaw **136**. The second rear catching outer jaw **136** may be formed on the rear side of the hollow cover **120** in the diagonal direction. The second rear catching outer jaw **136** may be disposed on the upper and lower sides at the left and right sides of the second front catching outer jaw **135**.
- (44) Referring to FIGS. 8 and 9, when the hollow cover **120** is inserted and rotated into the mounting part **7**, the second detachable part **134** may be detached from the first detachable part **131**.
- (45) The detachable part **130** can simply assemble the fastening module **10** which is a pre-prepared tolerance module, and thus workability can be improved, and performance improvement such as quality and cost can be expected.
- (46) Referring to FIGS. 5 and 6, the screw bushing part **100** may include a screw bush **150**. An

outer screw **151** may be formed on an outer circumferential surface of the screw bush **150**. The outer screw **151** of the screw bush **150** may be fastened to the inner screw **121** of the hollow cover **120**. As the screw bush **150** rotates, the screw bush **150** may linearly move with respect to the hollow cover **120**. The inner screw **121** of the hollow cover **120** and the outer screw **151** of the screw bush **150** may freely move linearly due to a large angle between them. The screw bush **150** may be fastened and inserted from the front to the rear of the hollow cover **120**. The screw bush **150** may be provided with a catching jaw **153**. The catching jaw **153** may be formed as a flange at a front surface of the screw bush **150**. The catching jaw **153** may be caught on a front surface of the hollow cover **120**.

(47) Referring to FIG. **10**, the screw bush **150** may include an one-direction slope catching jaw **152**. The one-direction slope catching jaw **152** may be formed inside the screw bush **150**. The one-direction slope catching jaws **152** may be formed to face each other at upper and lower sides. The one-direction slope catching jaw **152** may allow a front movement of a bolt part **200** and suppress a rear movement of the bolt part **200**. The one-direction slope catching jaw **152** may protrude to be inclined downward as it goes from an inner circumferential surface to a front of the screw bush **150**. A tip of the one-direction slope catching jaw **152** may have a concave arc shape. The concave arc shape is fitted to an outer circumferential surface of the bolt part **200** and can increase an anti-rear slip force.

(48) The one-direction slope catching jaw **152** enables an anti-rear slip of the bolt part **200** (a bolt thread and the catching jaw act like a hook to easily move to the front and be caught on the rear), maintenance of the center origin between the screw bush **150** and the bolt part **200** (securing XZ reference point for convenient tool use), prevention of flow between the screw bush **150** and the bolt part **200** (improvement of noise-causing factors such as hitting sound), and tolerance compensation by adjusting the XZ direction, thereby providing a flexible structure.

(49) Referring to FIGS. **5** and **6**, the fastening module **10** may include the bolt part **200**. The bolt part **200** may be fitted toward the front from the rear of the screw bush **150**.

(50) The bolt part **200** may include a washer based bolt **200**. The washer based bolt **200** may include a hexagonal head **210**, a male screw rod **220**, and a head washer **230** formed on the hexagonal head **210**. The hexagonal head **210** may be a tool application portion, the male screw rod **220** may be a fastening portion of a nut part **300**, and the head washer **230** may be a rear end pressing portion of the screw bush **150**. A tip **221** of the male screw rod **220** may be compressed after the fastening module **10** is assembled. The compression can prevent separation of the nut part **300** fastened to the male screw rod **220**.

(51) Referring to FIGS. **5** and **6**, the fastening module **10** may include the nut part **300**. Referring to FIGS. **7** to **10**, the nut part **300** may be fastened to the male screw rod **220** protruding forward from the front surface of the hollow cover **120**. Thus, the nut part **300** may pass from the inside to the outside of the vehicle body part **30**, and then closely couple inner and outer surfaces of the vehicle body part **30** together with the hollow cover **120** by a pressurization rotation of the bolt part **200**.

(52) The nut part **300** may include a washer based nut **300**. The washer based nut **300** may include a nut **310**. The nut **310** may be fastened to the male screw rod **220**. The washer based nut **300** may include a washer **330**. The washer **330** may be integrally formed with the nut **310**. The washer **330** may have an angled shape (polygonal shape). For example, the washer **330** may have a quadrilateral shape. The quadrilateral washer **330** is easily caught on the outer surface of the vehicle body part **30** and can prevent the nut **310** from idling.

(53) The vehicle body assembly structure according to another embodiment of the present disclosure may include the vehicle body part **30**.

(54) Referring to FIG. **4**, the vehicle body part **30** may include a vehicle body outer plate **40**. The vehicle body outer plate **40** may be an outer plate with a curved surface. The vehicle body outer plate **40** may include a grip withdrawing hole **44**. The grip **4** may be disposed inside the grip withdrawing hole **44**. An inner horizontal bent piece **45** may be formed at an edge of the grip

withdrawing hole **44**. The inner horizontal bent piece **45** may be horizontally bent inward. A guide protrusion **9** may be fitted to the outside of the inner horizontal bent piece **45**. Referring to FIG. **2**, the six guide protrusions **9** may be formed on the front surface of the housing **5** along a perimeter of the grip **4**. For example, the total of six guide protrusions **9** may include two upper, two lower, and left and right guide protrusions.

(55) The vehicle body part **30** may include a vehicle body inner plate **50**. The vehicle body inner plate **50** may be installed to be spaced apart from the vehicle body outer plate **40**.

(56) The vehicle body inner plate **50** may include an insertion part **51**.

(57) The insertion part **51** may include an insertion hole **52**. The insertion hole **52** may be a hole that penetrates back and forth. The insertion hole **52** may have an enough size to insert the quadrilateral washer **330**.

(58) The insertion part **51** may include a slot **53**. The slot **53** may have a key groove shape. The slot **53** may communicate with one side of the insertion hole **52**. The slot **53** may be a movement path of the male screw rod **220**. The inner plate **50** around the slot **53** may be caught and pressed by the quadrilateral washer **330**.

(59) The vehicle body inner plate **50** may include an anti-rotation piece **54**. The anti-rotation piece **54** may be a plate bent 90 degrees after pressing the vehicle body inner plate **50** in a $\sqsubset$ -shape. The anti-rotation piece **54** may be disposed on or below the slot **53**. One side of the quadrilateral washer **330** may be adjacent to a lower surface of the anti-rotation piece **54**. The anti-rotation piece **54** is caught by the quadrilateral washer **330** and thus can prevent idling of the nut **310**.

(60) The vehicle body inner plate **50** may include a part insertion hole **55**. The front surface of the housing **5** of the part **3** may be inserted and disposed in the part insertion hole **55**. In this case, the mounting part **7** may be disposed inside the vehicle body inner plate **50**.

(61) FIGS. **11** to **16** illustrate order of assembling the parts of FIG. **1** to the vehicle body.

(62) More specifically, FIG. **11** illustrates before the nut part **300** of the part unit **1** is inserted into the insertion hole **52** of the vehicle body inner plate **50**.

(63) FIG. **12** illustrates after the nut part **300** of the part unit **1** is inserted into the insertion hole **52** of the vehicle body inner plate **50**.

(64) Referring to FIG. **13**, when the nut part **300** is inserted into the insertion hole **52**, the part **3** is disposed at a mounting position by moving the male screw rod **220** toward the slot **53**.

(65) Referring to FIG. **14**, when the housing **5** is pushed forward and inserted after the part **3** moves and is disposed at the mounting position, as illustrated in FIG. **32**, the 6-point guide protrusion **9** is placed outside the inner horizontal bent piece **45**, and the grip **4** is precisely disposed in the grip withdrawing hole **44**. Further, the front surface of the housing **5** is inserted into the part insertion hole **55**, and a guide cover **70** is matched and inserted into an inner peripheral surface of the inner horizontal bent piece **45**.

(66) When the guide protrusion **9** is placed outside the inner horizontal bent piece **45**, as illustrated in FIG. **15**, the bolt part **200** and the screw bush **150** advance forward by a force of the tool pressing the bolt part **200**, i.e., a force of the head washer **230** pressing a rear end of the screw bush **150**. The screw bush **150** can freely move at a large screw angle to automatically compensate as much as a gap with the vehicle body inner plate **50**.

(67) Referring to FIG. **16**, when the tool continues to operate and the bolt part **200** rotates in place in a state where the front surface of the screw bush **150** contacts an inner surface of the vehicle body inner plate **50**, the nut part **300** does not rotate by the anti-rotation piece **54** and is retracted along the male screw rod **220**. The quadrilateral washer **330** of the nut part **300** is pressed and coupled together with the screw bush **150** while being caught on an outer surface of the vehicle body inner plate **50**. The screw bush **150** may serve as a chock between the bolt part **200** and the nut part **300**.

(68) FIG. **17** is a separation perspective view of a vehicle body assembly structure according to another embodiment of the present disclosure when viewed from the rear.

(69) Referring to FIG. 17, a fastening module **10'** and a vehicle body inner plate **50'** according to another embodiment are similar to the fastening module **10** and the vehicle body inner plate **50** according to an embodiment. On the other hand, a nut part **300'** of the fastening module **10'** and an insertion part **51'** of the vehicle body inner plate **50'** are different from the nut part **300** of the fastening module **10** and the insertion part **51** of the vehicle body inner plate **50** in a shape and a function.

(70) FIG. 18 is a perspective view illustrating the fastening module **10'** of FIG. 17.

(71) Referring to FIG. 18, the nut part **300'** of the fastening module **10'** may include a shape modification nut part **300'**.

(72) The shape modification nut part **300'** may include a front ring **310'** and a rear ring **320'** fitted to a male screw rod **220**. The shape modification nut part **300'** may consist of a nut in which a female thread (not shown) is formed on an inner circumferential surface of the front ring **310'**.

(73) The shape modification nut part **300'** may include a plurality of bridges **330'**. The plurality of bridges **330'** may have a band shape. The plurality of bridges **330'** may connect the front ring **310'** and the rear ring **320'** along a circumference between the front ring **310'** and the rear ring **320'**. The bridge **330'** may have a decreasing thickness as it goes toward the center between the front ring **310'** and the rear ring **320'**. That is, a lower surface of the bridge **330'** may have an arch shape. The center of the bridge **330'** is configured to have a thin thickness, and thus the bridges **330'** can be folded well while the front ring **310'** moves toward the rear ring **320'**.

(74) The shape modification nut part **300'** may include an anti-slip protrusion **340'**. The anti-slip protrusion **340'** may be caught on an outer surface of the vehicle body inner plate **50'**. The anti-slip protrusion **340'** may include an inclined surface **341'**. The inclined surface **341'** may have a slope with an increasing inclination as it goes to the rear. The inclined surface **341'** may provide a flexible insertion of the insertion part **51'**. The anti-slip protrusion **340'** may be formed on an outer surface of the rear ring **320'** or a rear outer surface of the bridge **330'**.

(75) Referring to FIG. 17, the vehicle body inner plate **50'** may include the insertion part **51'**. The insertion part **51'** may include an insertion hole **52'**. The insertion hole **52'** may have an enough size to insert the shape modification nut part **300'**. A diameter of the insertion hole **52'** may be less than an outer diameter of the anti-slip protrusion **340'**. That is, the anti-slip protrusion **340'** can be forcibly inserted into the insertion hole **52'** by using the inclined surface **341'** and elastic deformation.

(76) FIGS. 19 to 22 illustrate order of assembling parts of FIG. 17 to a vehicle body.

(77) More specifically, FIG. 19 illustrates before the shape modification nut part **300'** of the part unit **1** is inserted into the insertion hole **52'** of the vehicle body inner plate **50'**.

(78) FIG. 20 illustrates after the shape modification nut part **300'** of the part unit **1** is inserted into the insertion hole **52'** of the vehicle body inner plate **50'**. That is, when the shape modification nut part **300'** is inserted into the insertion hole **52'** of the vehicle body inner plate **50'**, a rear surface of the anti-slip protrusion **340'** is caught on the outer surface of the vehicle body inner plate **50'** to the extent that it does not slip.

(79) When the shape modification nut part **300'** is inserted into the insertion hole **52'**, as illustrated in FIG. 32, the 6-point guide protrusion **9** is placed outside the inner horizontal bent piece **45**, and the grip **4** is precisely disposed in the grip withdrawing hole **44**. Further, the front surface of the housing **5** is inserted into the part insertion hole **55**, and the guide cover **70** is matched and inserted into the inner peripheral surface of the inner horizontal bent piece **45**.

(80) When the guide protrusion **9** is placed outside the inner horizontal bent piece **45**, as illustrated in FIG. 21, the bolt part **200** and the screw bush **150** advance forward by a force of the tool pressing the bolt part **200**, i.e., a force of the head washer **230** pressing the rear end of the screw bush **150**. The screw bush **150** can freely move at a large screw angle to automatically compensate as much as a gap with the vehicle body inner plate **50'**.

(81) Referring to FIG. 22, when the tool continues to operate and the bolt part **200** rotates in place

in a state where the anti-slip protrusion **340'** of the shape modification nut part **300'** is caught on the outer surface of the vehicle body inner plate **50'** and does not rotate by an anti-rotation piece **54'** in a state where the front surface of the screw bush **150** contacts an inner surface of the vehicle body inner plate **50'**, the front ring **310'** is retracted along the male screw rod **220**. Hence, as the center of the bridge **330'** is folded and spread, it is pressed and coupled together with the screw bush **150** while being caught on the outer surface of the vehicle body inner plate **50'**. The screw bush **150** may serve as a chock between the bolt part **200** and the shape modification nut part **300'**.

(82) FIG. **23** is a separation perspective view of a vehicle body assembly structure according to yet another embodiment of the present disclosure when viewed from the rear.

(83) Referring to FIG. **23**, a fastening module **10''** according to yet another embodiment is similar to the fastening module **10'** according to another embodiment, but a nut part **300''** of the fastening module **10''** is different from the nut part **300'** of the fastening module **10'** in a shape and a function.

(84) FIG. **24** is a perspective view illustrating the fastening module **10''** of FIG. **23**.

(85) Referring to FIG. **24**, the nut part **300''** of the fastening module **10''** may include a shape modification nut part **300''**.

(86) The shape modification nut part **300''** may include a nut **310''**. The nut **310''** may have a ring shape. The nut **310''** may be fastened to a male screw rod **220**.

(87) The shape modification nut part **300''** may include a plurality of inclined wings **320''**. The inclined wing **320''** may protrude to be upward inclined as it goes toward the rear from the nut **310''**. That is, the inclined wing **320''** may be a wedge type. The inclined wing **320''** may have a shape that can be well bent and spread. An outer diameter of a tip of the inclined wing **320''** may be much greater than an inner diameter of an insertion hole **52'**.

(88) FIGS. **25** to **28** illustrate order of assembling parts of FIG. **23** to a vehicle body.

(89) More specifically, FIG. **25** illustrates before the shape modification nut part **300''** of the part unit **1** is inserted into an insertion hole **52'** of a vehicle body inner plate **50'**.

(90) FIG. **26** illustrates after the shape modification nut part **300''** of the part unit **1** is inserted into the insertion hole **52'** of the vehicle body inner plate **50'**. That is, when the shape modification nut part **300''** is inserted into the insertion hole **52'** of the vehicle body inner plate **50'**, the inclined wing **320''** is caught on an outer surface of the vehicle body inner plate **50'** while the inclined wing **320''** is elastically deformed downward and then is restored.

(91) When the shape modification nut part **300''** is inserted into the insertion hole **52'**, as illustrated in FIG. **32**, the 6-point guide protrusion **9** is placed outside the inner horizontal bent piece **45**, and the grip **4** is precisely disposed in the grip withdrawing hole **44**. Further, the front surface of the housing **5** is inserted into the part insertion hole **55**, and the guide cover **70** is matched and inserted into the inner peripheral surface of the inner horizontal bent piece **45**.

(92) When the guide protrusion **9** is placed outside the inner horizontal bent piece **45**, as illustrated in FIG. **27**, the bolt part **200** and the screw bush **150** advance forward by a force of the tool pressing the bolt part **200**, i.e., a force of the head washer **230** pressing the rear end of the screw bush **150**. The screw bush **150** can freely move at a large screw angle to automatically compensate as much as a gap with the vehicle body inner plate **50'**.

(93) Referring to FIG. **28**, when the tool continues to operate in a state where the inclined wing **320''** of the shape modification nut part **300''** is caught on the outer surface of the vehicle body inner plate **50'** and does not rotate by an anti-rotation piece **54'** in a state where the front surface of the screw bush **150** contacts the inner surface of the vehicle body inner plate **50'**, the bolt part **200** rotates in place, and the nut **310''** is retracted along the male screw rod **220**. Hence, as the inclined wing **320''** is folded and spread, the inclined wing **320''** is pressed and coupled together with the screw bush **150** while being caught on the outer surface of the vehicle body inner plate **50'**. The screw bush **150** may serve as a chock between the bolt part **200** and the shape modification nut part **300''**.

(94) FIGS. **29** and **30** are perspective views illustrating before and after a guide cover is assembled

to a part. FIG. 31 is a perspective view illustrating coupling parts of FIG. 30 to a vehicle body.

(95) Referring to FIGS. 29 and 30, the vehicle body assembly structure according to an embodiment of the present disclosure may include the guide cover 70. The guide cover 70 may cover a grip 4 of a flush handle. The guide cover 70 may be matched to the grip withdrawing hole 44. A front surface of the guide cover 70 may protrude outward from the grip withdrawing hole 44.

(96) The guide cover 70 can secure skin consistency. That is,

(97) (1) Due to characteristics of the flush handle, it is important to secure the design consistency between the vehicle body outer plate 40 and the grip 4 on the outside of the vehicle (match of skin surface/match of gap without leaning of the product).

(98) (2) Precisely seated in the grip withdrawing hole 44 using the 6-point guide protrusion 9 and a double guide structure capable of additionally securing a correct position.

(99) (3) Complete by stably fastening the fastening modules 10, 10', and 10'' with a tolerance structure in a state of being double guided by the 6-point guide protrusion 9 and the guide cover 70, and then removing the guide cover 70 (a detachable jig of the guide cover is recyclable).

(100) Some embodiments or other embodiments of the present disclosure described above are not mutually exclusive or distinct from each other. Configurations or functions of some embodiments or other embodiments of the present disclosure described above can be used together or combined with each other.

(101) It is obvious to those skilled in the art that the present disclosure can be embodied in other specific forms without departing from the spirit and essential features of the present disclosure. Accordingly, the above detailed description should not be construed as limiting in all aspects and should be considered as illustrative. The scope of the present disclosure should be determined by rational interpretation of the appended claims, and all modifications within an equivalent scope of the present disclosure are included in the scope of the present disclosure.

Claims

1. A part unit assembled to a vehicle body part, comprising: a part on which a plurality of mounting parts is formed; and a fastening module assembled to each of the plurality of mounting parts, wherein the fastening module includes: a screw bushing part installed on the mounting part to be linearly movable back and forth; a bolt part fitted toward a front from a rear of the screw bushing part; and a nut part fastened to a male screw rod of the bolt part protruding forward, wherein the nut part passes from an inside to an outside of the vehicle body part, and then the nut part and the screw bushing part closely couple inner and outer surfaces of the vehicle body part by a pressurization rotation of the bolt part, wherein the screw bushing part includes: a screw cover part installed on the mounting part; and a screw bush screw-fastened to the screw cover part, wherein the screw bush linearly moves forward by the pressurization rotation of the bolt part.
2. The part unit of claim 1, wherein the nut part includes washer based nut.
3. The part unit of claim 1, wherein the screw cover part includes: a hollow cover in which a screw is formed on an inner circumferential surface of the hollow cover; and a detachable part configured to detach the hollow cover from the mounting part.
4. The part unit of claim 3, wherein the detachable part includes: a first detachable part formed on the mounting part; and a second detachable part formed on the hollow cover, wherein when the hollow cover is inserted and rotated into the mounting part, the second detachable part is detached from the first detachable part.
5. The part unit of claim 4, wherein the first detachable part includes: a first front catching inner jaw formed at front left and right sides of the mounting part; and a first rear catching inner jaw formed at a diagonal rear of the mounting part, wherein the second detachable part includes: a second front catching outer jaw formed at front left and right sides of the hollow cover; and a second rear catching outer jaw formed at a diagonal rear of the hollow cover, wherein the second

front catching outer jaw is caught on the first front catching inner jaw, and wherein the second rear catching outer jaw is caught on the first rear catching inner jaw.

6. The part unit of claim 1, wherein facing one-direction slope catching jaws are formed inside the screw bush, and wherein the one-direction slope catching jaws allow a front movement of the bolt part and suppress a rear movement of the bolt part.

7. The part unit of claim 6, wherein the one-direction slope catching jaws protrude to be inclined downward as they go from an inner circumferential surface to a front of the screw bush.

8. The part unit of claim 1, wherein the bolt part includes a washer based bolt.

9. A vehicle body assembly structure in which a part unit is assembled to a vehicle body part, wherein the vehicle body part includes a vehicle body outer plate and a vehicle body inner plate, wherein the vehicle body inner plate includes: an insertion part penetrated back and forth; and an anti-rotation piece formed on an outer surface of the vehicle body inner plate, wherein the part unit includes: a part on which a plurality of mounting parts is formed; and a fastening module configured to fasten and couple the plurality of mounting parts to the vehicle body inner plate, wherein the fastening module includes: a screw bushing part installed on the mounting part to be linearly movable back and forth; a bolt part fitted toward a front from a rear of the screw bushing part; and a nut part fastened to a male screw rod of the bolt part protruding forward, wherein the nut part is inserted into the insertion part and is disposed between the vehicle body outer plate and the vehicle body inner plate, and wherein the nut part and the screw bushing part closely couple inner and outer surfaces of the vehicle body inner plate by a rotation of the bolt part.

10. The vehicle body assembly structure of claim 9, wherein the insertion part includes: an insertion hole through which the nut part passes; and a slot configured to communicate with the insertion hole, wherein the slot is formed as a movement path of the male screw rod of the bolt part, wherein the nut part includes an angled washer based nut, wherein one side of the angled washer based nut is caught on the anti-rotation piece, and wherein one surface of the angled washer based nut is caught on the outer surface of the vehicle body inner plate and is pressed.

11. The vehicle body assembly structure of claim 9, wherein the screw bushing part includes: a screw cover part installed on the mounting part; and a screw bush screw-fastened to the screw cover part.

12. The vehicle body assembly structure of claim 11, wherein the screw cover part includes: a hollow cover in which a screw is formed on an inner circumferential surface of the hollow cover; and a detachable part configured to detach the hollow cover from the mounting part.

13. The vehicle body assembly structure of claim 12, wherein the detachable part includes: a first detachable part formed on the mounting part; and a second detachable part formed on the hollow cover, wherein when the hollow cover is inserted and rotated into the mounting part, the second detachable part is detached from the first detachable part.

14. The vehicle body assembly structure of claim 13, wherein the first detachable part includes: a first front catching inner jaw formed at front left and right sides of the mounting part; and a first rear catching inner jaw formed at a diagonal rear of the mounting part, wherein the second detachable part includes: a second front catching outer jaw formed at front left and right sides of the hollow cover; and a second rear catching outer jaw formed at a diagonal rear of the hollow cover, wherein the second front catching outer jaw is caught on the first front catching inner jaw, and wherein the second rear catching outer jaw is caught on the first rear catching inner jaw.

15. The vehicle body assembly structure of claim 12, wherein facing one-direction slope catching jaws are formed inside the screw bush, and wherein the one-direction slope catching jaws allow a front movement of the bolt part and suppress a rear movement of the bolt part.

16. The vehicle body assembly structure of claim 9, wherein the part includes a flush handle with a grip that is withdrawn, wherein a part insertion hole, in which a front side of the part is inserted and disposed, is formed in the vehicle body inner plate, and wherein a grip withdrawing hole, from which the grip is withdrawn, is formed in the vehicle body outer plate.

17. The vehicle body assembly structure of claim 16, wherein an inner horizontal bent piece, that is horizontally bent inward, is formed at an edge of the grip withdrawing hole, and wherein a guide protrusion fitted to an outside of the inner horizontal bent piece is formed on a front surface of the part.
18. The vehicle body assembly structure of claim 17, wherein a guide cover is detachably installed between the guide protrusion and the grip, and wherein the guide cover is matched to the grip withdrawing hole and is inserted into and protrudes from the grip withdrawing hole.
19. The vehicle body assembly structure of claim 9, wherein the bolt part includes a washer based bolt.
20. The part unit of claim 1, wherein the nut part includes a shape modification nut part.
21. The part unit of claim 20, wherein the shape modification nut part includes: a front ring and a rear ring fitted to the male screw rod; a plurality of bridges configured to connect the front ring and the rear ring; a female thread formed on an inner circumferential surface of the front ring; and an anti-slip protrusion formed at a rear of the bridges or an outer surface of the rear ring, wherein the bridges are folded and widely spread while the front ring moves toward the rear ring.
22. The part unit of claim 21, wherein each of the plurality of bridges has a decreasing thickness as it goes toward a center between the front ring and the rear ring.
23. The part unit of claim 20, wherein the shape modification nut part includes: a nut fastened to the male screw rod; and a plurality of inclined wings protruding to be upward inclined as they go toward a rear from the nut, wherein the inclined wings are bent and widely spread while the nut moves toward the vehicle body part.
24. The vehicle body assembly structure of claim 9, wherein the nut part includes a shape modification nut part, and wherein the insertion part an insertion hole through which the shape modification nut part passes.
25. The vehicle body assembly structure of claim 24, wherein the shape modification nut part includes: a front ring and a rear ring fitted to the male screw rod; a plurality of bridges configured to connect the front ring and the rear ring; a female thread formed on an inner circumferential surface of the front ring; and an anti-slip protrusion formed on an outer surface of the rear ring, wherein a side surface of the anti-slip protrusion is caught on the anti-rotation piece, wherein a rear surface of the anti-slip protrusion is caught on the outer surface of the vehicle body inner plate, and wherein the bridges are folded, widely spread, caught on the outer surface of the vehicle body inner plate, and pressed while the front ring moves toward the rear ring.
26. The vehicle body assembly structure of claim 25, wherein each of the plurality of bridges has a decreasing thickness as it goes toward a center between the front ring and the rear ring.
27. The vehicle body assembly structure of claim 24, wherein the shape modification nut part includes: a nut fastened to the male screw rod; and a plurality of inclined wings protruding to be upward inclined as they go toward a rear from the nut, wherein the plurality of inclined wings is bent, widely spread, and pressed while the nut moves toward the outer surface of the vehicle body inner plate with the plurality of inclined wings being caught on the outer surface of the vehicle body inner plate.
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